

A Practitioner's Guide to Agentic AI tools

From Chatbot to Autonomous Agents

Gonzalo Balletero

Penn State University

Applied and Quantitative Economics (ECON 533)

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Outline

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Why AI Agents?

Why AI Agents?

- Most of you use chatbots (e.g., ChatGPT, Claude.ai)
 - You know how useful they are, and probably their limits
- With chatbots, every new session starts from scratch. They forget how you work
- AI agents solve this, and take it much further

The chatbot is a consultant you call (and forgets everything between calls).

AI agents are a team of co-workers who assist you to get the work done.

AI Agents as Research Assistants

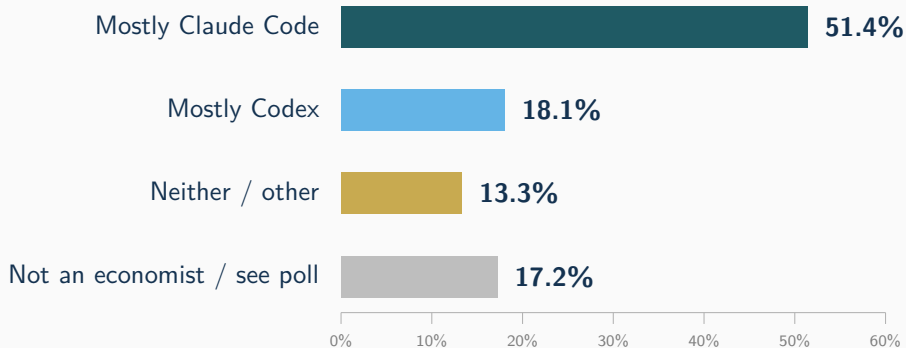
- AI agents can handle repetitive tasks that slow down your workflow (Korinek, 2025)
- What do AI agents do well?
 - Read data, run code, get estimates, and more ... (Grundl, 2026; Kohler et al., 2026)
- What do AI agents not do well (so far)?
 - Guarantee that the output they deliver is correct
 - Think as an economist (e.g., relevant questions, credible identification strategies)

A Shift in What Economists Need to Do

- AI produces output that looks *plausible* but can be wrong in ways that require *expertise to detect* (Andrews, 2026)
- The bottleneck is no longer *coding*, it is *evaluating* output (how and when)

How to Deploy AI Agents?

Which AI Agents Are Economists Using?



Source: @aniketapanjwani on X.com (459 votes, March 6, 2026).

Two Ways to Use Deploy Claude Code: CLI vs. IDE

1. Terminal (CLI)

- Run `cclaude` from any project folder in the terminal

2. Integrated Development Environment (IDE)

- Claude Code integrated with Visual Studio Code
- Easier to supervise what Claude is doing

Setting up Claude Code with VS Code

- Open Visual Studio Code and go to the Extensions panel
- Search for “*Claude Code*” and click Install
- Sign in with your Anthropic account
 - Pro plan (\$20/month) is required (check with your accountant)
- Open any project folder and launch the Claude Code panel

How to Use Claude Code?

Project Folder Structure

- A clean folder structure is the foundation

```
my-project/  
├── CLAUDE.md  
├── skills/  
├── data/  
│   ├── raw/  
│   └── cleaned/  
├── scripts/  
├── figures/  
├── tables/  
└── manuscript/
```

The CLAUDE.md File

- It is a short rulebook that Claude reads at the start of every session
- It contains your project overview and ground rules
- Keep it under ~ 150 lines. Beyond that, context quality degrades

What Are Skills?

- Skills are instruction files that teach Claude how to do a specific task
- Think of it as a guide for a new RA
- Written once, applied every time

Getting Started

Read CLAUDE.md and all files in skills/ carefully.

Please confirm you understand the project, the ground rules, and the output conventions before we start.

Best Practices and Resources

Best Practices

- Use plan mode
- Be specific about the task and output format
- Be polite, as impolite prompts may result in poor performance (Yin et al., 2024)
- Trust but verify

*These tools are new and evolving fast.
Nobody has “solved” how to use them best.*

Resources

- *Mini-series on Claude Code for applied economists* by Paul Goldsmith-Pinkham
- *Claude Code course on Youtube* by Aniket Panjwani
- *AI for professionals who don't code* by Claude Blattman
- *A comprehensive guide to AI-assisted academic workflows* by Pedro Sant'Anna
- *Agentic AI bootcamp* by Erkmen G. Aslim and Emily Beam
- *AI tools for business and economic research*, compiled by Mihail Velikov
- *Ongoing AI resources thread on X* by Jeremy Nguyen

Live Demo

- Let's open Claude Code in VS Code and see it in action
- Today we will:
 - Solve assignments 1 and 7
 - Build an academic website from scratch